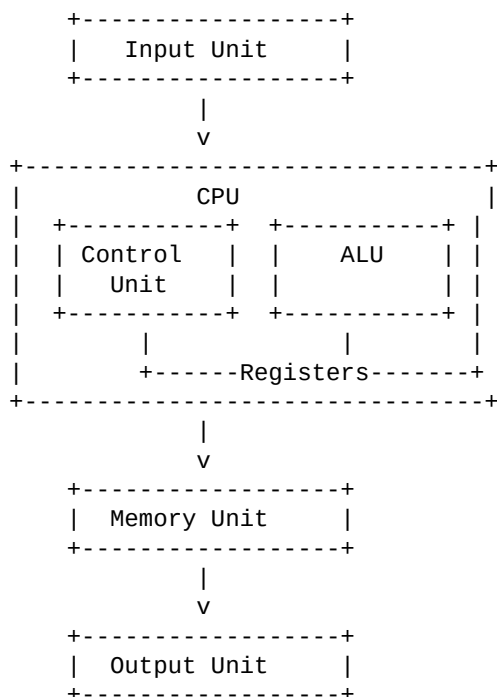


Q1. Draw and Explain the Block Diagram of a Computer System (10 Marks)

Block Diagram of Computer System



Explanation

A computer system is an electronic device that takes data as input, processes it according to given instructions, stores the data if required, and produces meaningful output. The basic block diagram of a computer system consists of the Input Unit, Central Processing Unit (CPU), Memory Unit, and Output Unit.

1. Input Unit

The input unit is responsible for accepting data and instructions from the user. It converts the input into a form that can be understood by the computer system. This unit acts as a link between the user and the computer. Common input devices include keyboard, mouse, scanner, and microphone.

2. Central Processing Unit (CPU)

The CPU is the most important part of the computer system and is often referred to as the brain of the computer. It performs all the processing operations and controls the working of all other units.

(a) Control Unit (CU)

The control unit manages and coordinates all activities of the computer system. It fetches instructions from memory, decodes them, and generates control signals. These control signals guide the ALU, memory, and input/output devices to perform the required operations in a proper sequence.

(b) Arithmetic Logic Unit (ALU)

The arithmetic logic unit performs all arithmetic and logical operations. Arithmetic operations include addition, subtraction, multiplication, and division, while logical operations include comparison, AND, OR, and NOT operations. The ALU receives data from registers and stores the result back in registers or memory.

(c) Registers

Registers are small and high-speed storage locations present inside the CPU. They temporarily store data, instructions, and intermediate results during processing. Because of their high speed, registers help in increasing the overall performance of the system.

3. Memory Unit

The memory unit stores data, instructions, and results of processing. It is divided into primary and secondary memory. Primary memory includes RAM and ROM. RAM is volatile and loses data when power is off, while ROM is non-volatile and stores permanent instructions.

4. Output Unit

The output unit provides the final processed information to the user in a human-readable form. It converts machine-level output into understandable format. Common output devices are monitor, printer, and speakers.

Q2. Explain the Following (10 Marks)

(a) Vector Processing

Vector processing is a processing technique in which a single instruction is applied to multiple data elements arranged in the form of vectors. In this method, operations are performed on entire arrays instead of individual elements.

This type of processing is mainly used in scientific and engineering applications where large sets of data need to be processed repeatedly. Vector processing reduces the number of instructions required and significantly improves execution speed.

Advantages: It increases computational speed by processing multiple data items simultaneously. It reduces instruction overhead. It is highly efficient for mathematical and scientific computations.

(b) Parallel Processing

Parallel processing is a technique in which multiple processors or processing units work simultaneously to execute different parts of a program. The main objective of parallel processing is to reduce execution time and increase system performance.

In this approach, a complex task is divided into smaller sub-tasks, and each sub-task is processed at the same time by different processors. Parallel processing is widely used in high-performance computing systems and servers.

Advantages: It improves processing speed and efficiency. It allows better utilization of system resources. It supports handling of large and complex problems.

(c) Pipelining

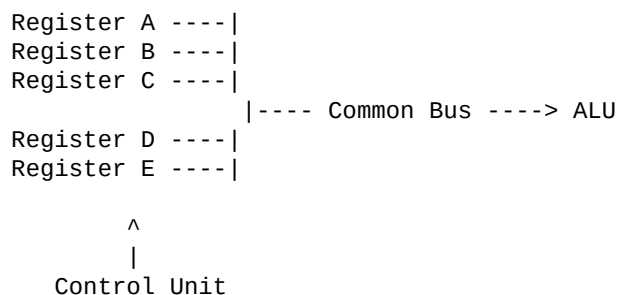
Pipelining is an instruction execution technique in which multiple instructions are overlapped during execution. The execution of an instruction is divided into different stages, and each stage processes a different instruction at the same time.

For example, while one instruction is being executed, another instruction can be decoded, and a third instruction can be fetched. This increases the instruction throughput of the CPU.

Advantages: It increases CPU performance. It ensures better utilization of processing units. It reduces instruction execution time.

Q3. Draw and Explain Interconnection Diagram of Common Bus System (10 Marks)

Interconnection Diagram



Explanation

A common bus system is used to transfer data between various registers using a single shared communication line called a bus. Instead of providing separate connections between each register, a common bus is used to reduce hardware complexity.

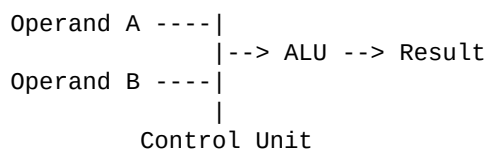
The control unit generates control signals that determine which register will place data on the bus and which register will receive data. At any given time, only one register is allowed to send data on the bus to avoid conflicts.

Advantages: It reduces the number of connecting wires. It simplifies system design. It is cost-effective.

Disadvantages: Only one data transfer can occur at a time. It may cause delay due to bus contention.

Q4. Explain Arithmetic Operations with the Help of Diagram (10 Marks)

Block Diagram of Arithmetic Operations



Explanation

Arithmetic operations are performed by the Arithmetic Logic Unit (ALU) of the CPU. These operations are essential for all computational tasks in a computer system.

1. Addition

Addition is performed using binary adders. It involves adding two binary numbers and generating a sum and carry if required.

2. Subtraction

Subtraction is performed using the two's complement method. In this method, subtraction is converted into addition to simplify hardware implementation.

3. Multiplication

Multiplication is carried out by repeated addition and shifting operations. The ALU uses shift registers to perform fast multiplication.

4. Division

Division is performed using repeated subtraction or shift-based methods. It produces both quotient and remainder.

Conclusion

Arithmetic operations form the foundation of all computing processes and are extensively used in scientific, engineering, and commercial applications.